

SACRAMENTO INTL, CA, USA

WMO: 724839

Lat: 38.696N

Lon: 121.590W

Elev: 23

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.8	33.6	20.7	15.5	50.8	24.8	18.9	50.8	29.5	54.3	26.5	54.2	3.0	320	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.6	100.3	70.1	97.5	69.1	94.6	67.8	72.6	95.7	71.0	93.4	69.4	90.9	6.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	88.8	84.1	62.4	84.5	82.3	60.8	79.8	79.8	36.2	95.4	34.6	93.7	33.4	91.0	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.8	19.7	17.3	DB	26.8	105.9	3.5	2.2	24.3	107.5	22.3	108.8	20.3	110.0	17.8	111.6	(4)
(5)			WB	25.8	75.5	3.7	2.4	23.1	77.2	20.9	78.6	18.8	79.9	16.1	81.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	47.5	50.8	55.3	59.4	66.5	73.1	76.8	75.8	72.4	63.9	53.6	47.3	(6)	
	DBStd	11.84	4.74	4.61	5.21	6.01	6.28	5.95	4.91	4.61	5.09	5.36	5.57	5.38	(7)	
	HDD50	309	105	42	13	3	0	0	0	0	0	0	27	119	(8)	
	HDD65	2475	541	398	304	187	56	6	0	0	6	85	342	550	(9)	
	CDD50	4657	28	64	175	285	511	693	830	800	671	431	135	34	(10)	
	CDD65	1348	0	0	1	18	102	249	365	335	227	50	1	0	(11)	
	CDH74	17645	0	0	63	406	1520	3280	4557	4080	2854	854	31	0	(12)	
	CDH80	9195	0	0	4	118	661	1737	2630	2295	1461	288	1	0	(13)	
(14) Wind	WSAvg	7.5	6.5	7.6	7.7	8.2	8.6	8.7	8.6	8.1	7.0	6.7	6.0	6.9	(14)	
(15) Precipitation	PrecAvg	17.6	2.9	2.9	2.4	1.3	0.6	0.2	0.0	0.0	0.1	0.9	1.7	3.9	(15)	
	PrecMax	24.4	8.7	6.2	7.4	3.7	1.6	1.4	0.0	0.7	0.6	3.5	3.9	9.4	(16)	
	PrecMin	6.2	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	5.0	2.2	1.8	2.0	1.0	0.5	0.4	0.0	0.2	0.2	1.1	0.9	2.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	71.4	79.7	89.0	97.1	103.1	104.7	103.7	99.7	93.0	78.0	65.1	(19)	
		MCWB	53.1	57.3	62.1	63.5	68.5	71.0	72.2	70.0	68.2	64.5	59.5	55.5	(20)	
	2%	DB	61.9	66.8	75.1	83.3	91.9	98.8	100.3	99.4	96.3	86.8	73.1	61.5	(21)	
		MCWB	53.4	56.5	59.6	61.1	65.7	69.7	70.8	69.8	66.9	62.5	58.2	54.4	(22)	
	5%	DB	58.8	63.3	71.8	78.9	87.8	94.3	97.1	95.7	92.6	82.5	68.7	59.0	(23)	
		MCWB	52.5	54.5	58.0	60.0	63.8	67.5	69.7	68.4	66.0	60.7	56.1	52.9	(24)	
	10%	DB	56.0	60.3	67.5	73.6	82.4	90.1	93.3	91.9	89.3	78.8	64.2	56.6	(25)	
		MCWB	51.2	53.3	56.1	58.1	62.2	66.4	68.6	67.3	65.0	59.3	55.0	51.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.0	60.8	64.9	66.5	70.9	74.0	75.4	74.1	71.8	67.8	63.1	59.6	(27)	
		MCDB	62.1	66.5	75.7	82.9	90.2	97.0	99.4	98.3	94.1	86.9	74.1	62.6	(28)	
	2%	WB	55.9	58.6	61.7	63.6	68.0	71.4	73.0	71.3	69.0	64.8	60.4	57.0	(29)	
		MCDB	59.2	64.3	72.0	78.8	88.3	94.1	96.1	95.4	90.7	81.8	68.8	59.9	(30)	
	5%	WB	53.7	56.3	59.4	61.5	65.8	69.4	71.2	69.6	67.4	62.7	58.5	54.8	(31)	
		MCDB	56.4	61.4	68.7	75.5	84.8	90.4	93.6	92.0	88.7	78.1	65.4	57.2	(32)	
	10%	WB	52.1	54.3	57.3	59.6	63.5	67.6	69.5	68.3	65.9	60.9	56.6	52.6	(33)	
		MCDB	55.2	58.7	65.3	72.0	80.4	87.8	90.8	89.7	86.3	74.8	62.7	55.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	17.8	21.3	24.3	27.1	30.3	32.6	32.5	31.3	27.9	20.7	16.0	(35)	
		MCDBR	21.6	23.7	29.2	33.1	34.9	36.1	36.1	36.8	36.8	34.9	27.9	19.1	(36)	
		MCWBR	13.8	14.5	15.8	15.1	13.7	13.3	13.7	13.2	13.4	14.7	14.6	12.1	(37)	
	5% WB	MCDBR	14.1	19.4	25.7	29.7	32.7	33.9	34.5	34.4	33.8	30.3	22.0	11.1	(38)	
		MCWBR	10.2	12.7	14.9	14.7	14.1	13.8	14.2	13.1	13.2	13.7	13.0	8.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.322	0.325	0.338	0.354	0.364	0.360	0.361	0.344	0.349	0.343	0.327	0.331	(40)	
	taud		2.511	2.509	2.461	2.405	2.402	2.420	2.406	2.446	2.442	2.468	2.500	2.479	(41)	
	Ebn at Noon		272	287	293	293	290	290	289	292	285	275	267	258	(42)	
	Edh at Noon		25	28	32	36	37	36	37	35	33	29	25	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		685	1014	1446	1923	2339	2611	2602	2327	1885	1326	846	610	(44)	
	RadStd		128	131	153	166	141	106	75	74	65	89	62	103	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-2.09	N/A	-1.03	-1.15	N/A	N/A	N/A	N/A	(46)
(47)	Regional (47 neighbors)	+0.72	N/A	-1.94	N/A	N/A	N/A	N/A	-209	+248	+76	(47)

Nomenclature: See separate page